Machine Learning - Detailed Guide with Examples & Diagrams

1. Introduction to Machine Learning

Machine Learning is a subset of AI that enables systems to learn from data.

Example: Email spam detection learns from past emails.

2. Types of Machine Learning

Supervised Learning: Uses labeled data (e.g., predicting house price).

Unsupervised Learning: Finds patterns (e.g., customer segmentation).

Reinforcement Learning: Learns via rewards (e.g., game playing AI).

3. Data Preprocessing

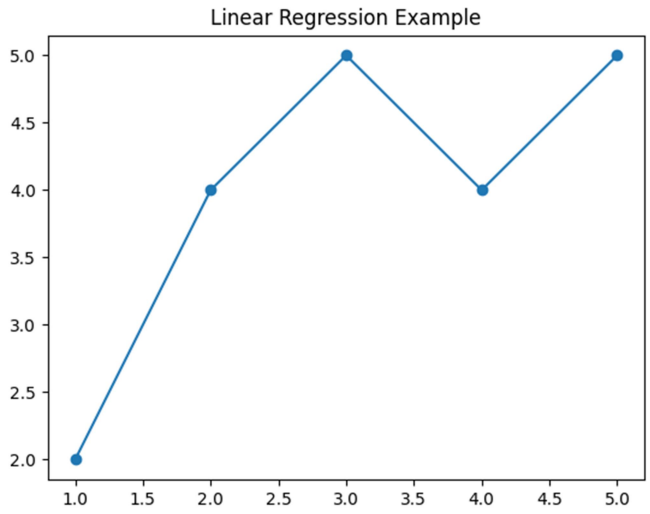
Cleaning data, handling missing values, scaling features.

Example: Filling missing age values in a dataset.

4. Regression

Used for predicting continuous values.

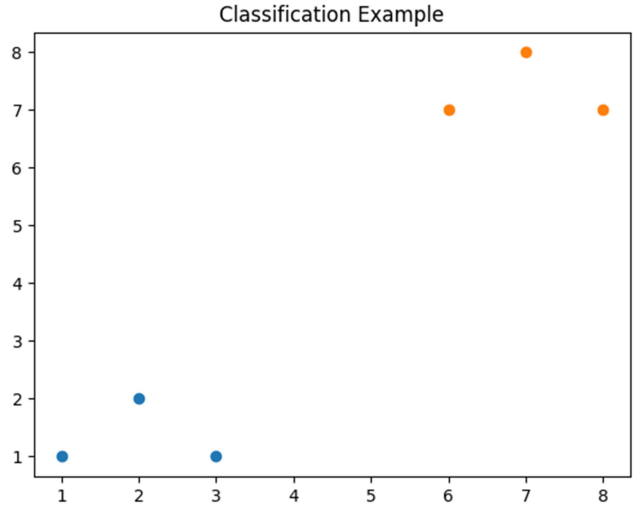
Example: Predicting house prices.



5. Classification

Used for categorical outputs.

Example: Spam vs Not Spam email classification.



6. Model Evaluation

Metrics: Accuracy, Precision, Recall.

Example: Checking how many predictions are correct.

7. Overfitting & Underfitting

Overfitting: Model memorizes data.

Underfitting: Model fails to learn patterns.

8. Basic Coding Example

```
import pandas as pd

from sklearn.linear_model import LinearRegression

model = LinearRegression()
```